SOKOINE UNIVERSITY OF AGRICULTURE

SCHOOL OF ENGINEERING AND TECHNOLOGY (SoET)

DEPARTMENT OF AGRICULTURAL ENGINEERING (DAE)

AE 219

ASSIGNMENT 4

**Note**:

[1] Please listen to videos https://www.youtube.com/playlist?list=PLmKnXyyknMkPhutqZ6uebPazrUCobJ8Ep provided by my Lab YEESI Lab and instructed by Dr Masamaki

[2] Please use Google Colab: https://colab.research.google.com/

[3] Please read and listen to the videos at https://towardsdatascience.com/best-time-saving-python-tricks-in-jupyter-notebook-fca877132507

[4] Please read how to input data into a program: https://www.askpython.com/python/examples/python-user-input

**Questions (All programs should run on the command line. Read Instruction 4)**:

1. **Write a Python program to process and display Wireless Sensor Network (WSN) installed at SUA Model Farm weekly. Use Python in-built data structures like a dictionary, list etc., to hold your data variables. Assume the program processes sensor data for one week only**
   - ***1.1.*** The program should accept names and locations (longitude and Latitude) for only 30 sensors {Name, location }. *The program should only take sensors located in the SUA model farm.*
   - ***1.2.*** The program should accept data ( $R_n$ net radiation at the crop surface [MJ $m^{-2}$ $day^{-1}$], G soil heat flux density [MJ $m^{-2}$ $day^{-1}$], T air temperature at 2 m height [°C], $u_2$ wind speed at 2 m height [m $s^{-1}$], $e_s$ saturation vapour pressure [kPa], $e_a$ actual vapour pressure [kPa], , $\gamma$ psychrometric constant [kPa $°C^{-1}$].) every hour every day for seven days, from Monday to Sunday. Sensor data should be collected in daylight only from 7:00 AM to 6:00 PM.
   - ***1.3.*** The program should calculate each sensor's daily ETo reference evapotranspiration [mm day-1]. See the formula for $\Delta$ slope vapour pressure curve [kPa $°C^{-1}$] below.

$$ET_o = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T+273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)}$$

$$\Delta = \frac{4098\left[0.6108\exp\left(\frac{17.27T}{T+237.3}\right)\right]}{(T+237.3)^2}$$

*1.4.* The program should display the names of sensors and daily reference evapotranspiration in descending order of the salary.

*1.5.* The program should be able to show/display the day in which most of the sensors had the largest reference evapotranspiration.

*1.6.* Note that >> The program should have a menu to allow the Model farm Technician to do the above five tasks

2. **Write a Python program to process and display Normalized Difference vegetation Index (NDVI) plant values of the Rice Farm located in SUA Model Farm. A technician chose 20 plants randomly on the rice farm and collected the Red and Near-infrared (NIR) reflectance signature using a Multispectral sensor on the plants' leaves to measure crop vigour for precision fertilization. Use Python in-built data structures like list or dict to hold your data variables. Assume the technician collected data every week for 16 weeks only.**

**2.1.** The program should accept plant heights, widths and locations (longitude and Latitude) for only 20 plants {height, width, location }. *The program should only take plants located on the SUA model rice farm.*

***2.2.*** The program should accept plant data (red and near-infrared reflectance) every week for 16 weeks.

***2.3.*** The program should calculate the weekly NDVI for each plant. Use the NDVI formula below:

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)}$$

***2.4.*** The program should display the locations, heights, widths and NDVI of the chosen plants in descending order of the NDVI.

***2.5.*** The program should be able to show/display the best day when most plants had high NDVI.

***2.6.*** Note that >> The program should have a menu to allow the technician to do the above five tasks

Submit your code results by Wednesday midnight (15-June-2022) to the CR as the softcopy and CR will forward the bundle of all codes to kadefue@sua.ac.tz. DO NOT SEND YOUR WORK to Instructor's email. Any enquiries, please send an email.

**As usual, Every one hour late equates to 5 marks deduction.**